

Advancing the Verification and Error Analysis of Thermohydraulic Simulations in Nuclear Reactors

Master Thesis

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Abstract

This internship report details the efforts to enhance the verification of advanced thermohydraulic simulation codes developed at CEA Saclay for the modeling of fluid flow dynamics in nuclear reactor systems. The focus was twofold: first, on refining the error evaluation scheme for 1D single-channels, and second, on introducing a new verification test case for two parallel channels with different inlet velocities. The project also involved significant improvements in the documentation of the verification database, transitioning it to a more user-friendly format using Jupyter Notebook.

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Chapter 1

General Introduction

1.1 Nuclear energy

The origins of nuclear energy can be traced back to the late 19th century, starting with Wilhelm Conrad Röntgen's discovery of X-rays. This was followed by Henri Becquerel's discovery of natural radioactivity. Pierre and Marie Curie, furthered our understanding by discovering highly radioactive elements: polonium and radium. The concept of artificial radioactivity was then introduced by Irène and Frédéric Joliot-Curie in 1934. [1]

A major breakthrough came in 1938 when Lise Meitner and Otto Frisch first demonstrated nuclear fission, which is splitting an atomic nucleus—in this case, uranium—into two smaller nuclei. This process was previously thought impossible and was found to release a large amount of energy along with additional neutrons. Scientists quickly realized that these neutrons could induce fission in other uranium atoms, leading to a self-sustaining series of reactions known as a **chain reaction**. Under controlled conditions, this chain reaction can be harnessed to generate power, whereas an uncontrolled reaction can result in an explosion of immense force. [2]

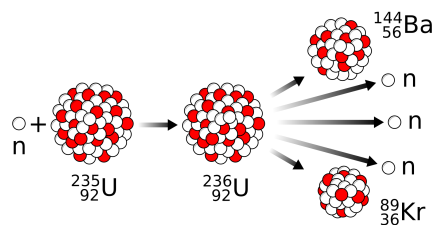


Figure 1.1: The fission reaction. A uranium U235 nucleus absorbs a neutron and splits into two lighter fragments. The process releases energy and new neutrons, which continues the chain reaction [10].

The potential of nuclear fission for both energy production and weapons quickly became clear. During World War II, the United States started the Manhattan Project, a large and secret effort to build an atomic bomb before Nazi Germany could. This project led to the creation of the atomic bombs that were dropped on Hiroshima and Nagasaki in Japan in 1945. These remain the only times nuclear weapons have been used in war.

After World War II, France established the Alternative Energies and Atomic Energy Commission (CEA) to focus on newly discovered nuclear energy [1]. Since then, France has become a global leader in nuclear power generation. For instance, all nuclear power plants in France utilize the Pressurized Water Reactor (PWR) design. As of 2022, approximately 63% of the country's electricity is generated from nuclear energy, making it the highest proportion globally. [4]

1.2 Overview of CEA

The French Alternative Energies and Atomic Energy Commission (CEA) is a public organism composed of 9 centers spread across all of France with about 22000 employees. It's split between four operational divisions tasked with carrying out its research:

- Energy Division, DES which focuses on low carbon energy production, management of resources and global performances.
- Military Applications Division, DAM which works on national security and defense.
- Technological Research Division, DRT tackles the big industrial and societal challenges of XXI century.
- Fundamental Research Division, DRF creates and produces new scientific knowledge and builds the foundation of fundamental research.

CEA is subdivided into various departments that work on diverse yet interconnected fields, including Defense and National Security, Technology for Future Medicine, Digital Technology, and Low-Carbon Energy (Nuclear and Renewable).

Specifically, the Department of Systems and Structural Modeling (DM2S), comprising three services:

- Service of Reactor Studies and Applied Mathematics (SERMA), which focuses on the neutronic aspects.
- Mechanical and Thermal Studies Service (SEMT), that studies materials used in the nuclear reactor.
- Thermohydraulics and Fluid Mechanics Service (STMF), which develops software used in studying the dynamics of fluid flow in the nuclear reactor.

This internship was conducted in STMF, in a lab where physics, math, computer science, and engineering are intertwined.

1.3 Nuclear power plant

A nuclear power plant is a type of thermal power station where the primary heat source is a nuclear reactor. Similar to conventional thermal power stations, nuclear power plants utilize heat to generate steam, which in turn drives a steam turbine connected to a generator for electricity production.

1.3.1 Pressurized Water Reactor (PWR)

The Pressurized Water Reactor (PWR) is a type of light-water nuclear reactor that was first designed for use in submarines. PWRs are now the most common type of nuclear reactor in power plants worldwide [12]. They generate heat through nuclear fission in the reactor core, which is then converted into electrical power.

Figure (1.2) illustrates the power transfer in a PWR. The orange path represents the primary coolant, which transports heat from the reactor core to the steam generator. The blue path represents the secondary coolant, which is converted into steam to drive the turbine and generate electricity.

Key Steps in the cycle of a PWR: [12]

- **Heat Generation:** Nuclear fission reactions occur in the reactor core, splitting uranium atoms in the fuel rods and generating heat.
- **Heat Transport:** The primary circuit uses high-pressure water, which avoids boiling due to its high pressure, to transport thermal energy to a steam generator.
- **Steam Production:** The heat from the primary circuit converts the water from the secondary circuit into steam.
- **Electricity Generation:** The steam drives a turbine, which spins an electric generator to produce electricity.

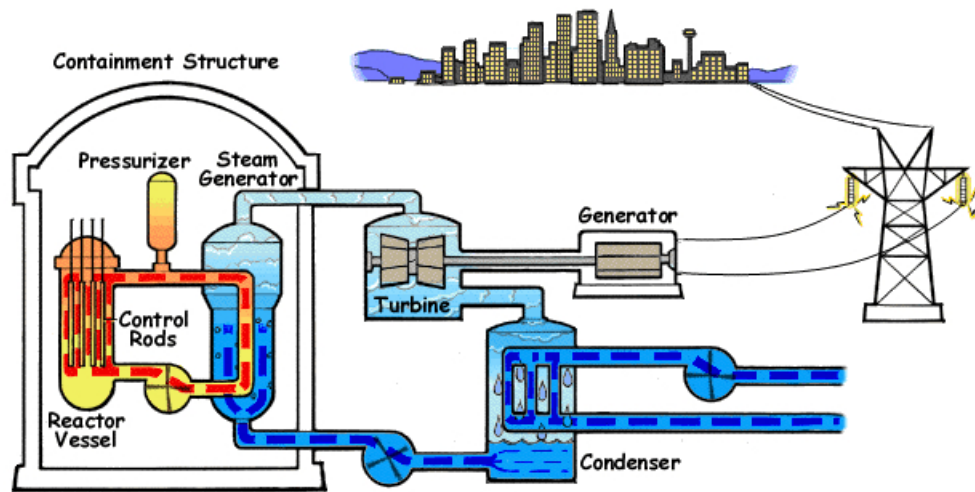


Figure 1.2: power transfer in a pressurized water reactor. (orange: Primary coolant; blue: the secondary coolant (steam and later feedwater)) [11].

To complete the cycle, a condenser converts steam back into liquid water, which is then returned to the steam generator using pumps. The condenser requires an external cooling source, such as a lake or sea, to cool the steam. In this process, the temperature of the cooling source is less critical than the speed at which its water is pumped into the condenser. It is important that the water flow rate is high enough to prevent a large quantity of heat from being transferred to the water at once, which could negatively impact the environment. This step is crucial as it allows the water to be reused in the cycle.

1.3.2 Safety in a nuclear reactor

Historically, the nuclear industry has witnessed several significant accidents. These include the **Three Mile Island** accident in 1979 that is considered the most serious nuclear accident in U.S. history; The **Chernobyl** disaster in 1986, regarded as the world's most devastating nuclear disaster to date; and most recently, the **Fukushima Daiichi** nuclear disaster in 2011, which was triggered by a tsunami that disabled the power supply and cooling mechanisms of three reactors at the facility. [15]

The high energy density in a nuclear reactor necessitates a significant emphasis on safety systems. Any failure in these systems can lead to situations requiring substantially higher energy for evacuation and containment. Each of the above incidents has played a role in the evolution of safety systems in nuclear reactors.

Many systems are put in place to ensure the safe operation of a nuclear reactor. We can categorize them under three main safety functions that should be continuously maintained: [9]

- **Control of the chain reaction:** Ensures that the nuclear fission process is kept within safe limits.
- **Cooling of the nuclear fuel:** When the reactor is in operation, it is necessary to continuously evacuate the heat generated by the nuclear fuel. When the reactor is stopped, it is also necessary to evacuate the residual heat that remains after the end of the chain reaction.
- **Confinement of radioactivity:** This involves preventing the dispersion of radioactive substances into the environment, and ensuring the protection of people and the environment against ionizing radiation. This is mainly done by the **Containment Structure** which is made of 3 layers, fuel cladding - reactor vessel - containment building (concrete).

Most safety systems in nuclear reactors are active, requiring intervention from operators or electric generators. However, there is a growing trend to incorporate more **passive safety systems** that rely on natural forces and phenomena such as gravity and pressure differences. [3]

1.4 Context of this internship

At CEA, several advanced 3D thermohydraulic simulation codes have been developed to model the fluid flow dynamics within the primary circuit of a nuclear reactor. These simulations often involve complex phenomena such as compressible viscous multiphase flows.

The primary focus of this internship is on the **verification** of these software tools, as opposed to their validation. It is important to clearly distinguish between these two concepts:

- **Verification** involves confirming that the simulation code correctly implements the intended mathematical model. In other words, it ensures that the simulation code is solving the equations correctly.
- **Validation** involves confirming that the simulation accurately represents the real-world system it is intended to model. This process ensures that the results obtained from the simulation are consistent with experimental data.

Currently, the verification database for these software tools includes cases for 1D, 2D, and 3D single- and multiphase flows. The focus of this internship is specifically on studying 1D single-phase flow inside the reactor vessel.

In a simplified view of the reactor core as shown in figure 1.3, coolant water enters from the cold leg, flows downward, hits the bottom of the reactor vessel, and then flows upward. Studying the flow is interesting because it can replicate accident conditions inside the reactor vessel, such as:

- Leakage of water: Results in a loss of density in the flow.
- Pressurizer breakdown: Leads to variations in pressure.
- Failure of reaction control: Causes excess temperature.
- Failure of a pump: Results in a loss of velocity.

The cases covered are divided into two categories: single-channel and two-channel flows.

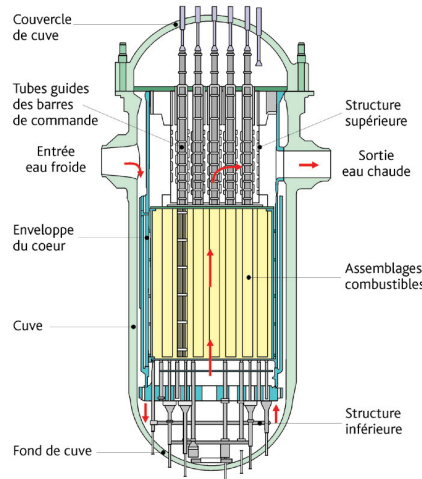


Figure 1.3: Vessel of a nuclear reactor [8].

1.4.1 Single channel: 1D single phase flows

The software currently contains a few verification tests for single-channel, single-phase cases such as: flow under gravity, regular friction, singular friction. The analytical solutions for these cases are derived under the assumption of incompressible flow and are then compared to the results of the software, which simulates a more general compressible flow.

The French Alternative Energies and Atomic Energy Commission (CEA) is a public organism composed of several research and development centers. At present, the success criteria are defined such that the relative error between the analytical and software

results should not exceed 0.01%. However, this criterion lacks scientific rigor, as it does not provide a comprehensive justification for its threshold. Our goal is to establish a scientifically-backed method to evaluate the errors for these cases. Additionally, we aim to improve the documentation and quality of these and other tests in the verification database by rewriting them using Jupyter Notebook. This approach allows for easy modification and automatic generation of PDF reports for the tests.

1.4.2 Two channels: 1D single phase flows

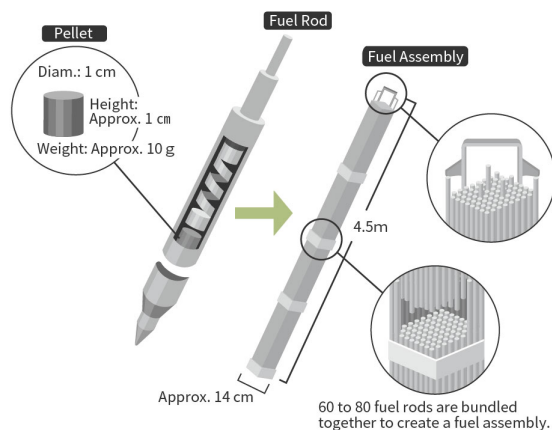


Figure 1.4: Fuel assembly [14].

The reactor vessel is typically composed of 150-250 fuel assemblies, each containing 200 to 300 fuel rods (see figure 1.4).

Flow dynamics: In the upward flow through the assemblies, differences in velocities can occur between the channels closest to the center assembly and the channels closest to the wall of the reactor. This is due to the geometry of the bottom vessel, and the different lengths of the “u-turn” that the water has to undertake while transitioning to the upward flow.

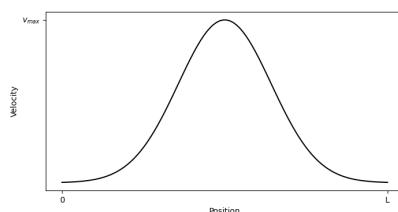


Figure 1.5: Initial velocity profile.

As shown in figure 1.5, the initial velocity profile at the start of the upward flow is not uniform. The velocity is highest at the center of the assembly and lower near the inner walls of the reactor core. This leads to the development of different flow regimes with varying initial velocities, resulting in **mixing**. These different flow regimes mix together and homogenize each other until their velocities become closer and the velocity profile flattens (figure 1.6). The **mixing length** is the distance at which the two flow regimes with different initial conditions (velocity in this case) blend together and homogenize. We are interested in studying this phenomenon.

This is important as the different upward flow regimes exert different upward forces (or lift forces) on the assemblies. These forces are not to be ignored! In fact, there is a spring system in place inside the

reactor vessel that holds the assembly and counteracts this lift force. One can notice the problem of having different lift forces being exerted on the center assembly and outer part of the assembly, which would require installing multiple different springs (different stiffness), which adds a lot of complexity in the design and especially in the refueling process. It's also important to note that this phenomenon happens under normal conditions of the reactor activity.

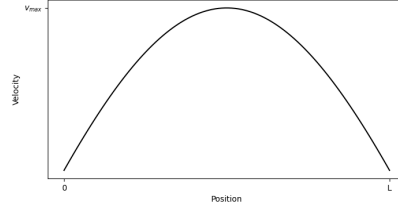


Figure 1.6: Flattened velocity profile.

Despite its importance, a scenario involving two parallel channels with differing initial conditions, specifically velocity, has not been thoroughly studied and is missing from the current verification database. Our objective is to incorporate this case into the verification database and ensure that the software can accurately capture the effects of these varying initial velocity. Specifically, we aim to see the "mixing length" effect, i.e the distance at which the flow from the two channels becomes homogeneous and indistinguishable.

1.4.3 Objectives of the Internship

In summary, the objectives of this internship are as follows:

- Propose a scientific error evaluation scheme for the 1D single-channel, single-phase cases.
- Add a new case involving 1D two-channel, single-phase flows to the verification database.
- Improve documentation by writing the test cases using Jupyter Notebook for better clarity and accessibility.

Chapter 2

Dynamics of single phase flow

The study of dynamic fluid flow is vast and complex. In this chapter, we focus on the aspects relevant to single-phase flow in nuclear reactors. We begin by presenting the fundamental equations governing the dynamics of single-phase fluid flows, particularly for a compressible viscous fluid. Then, we discuss the simplified incompressible model.

2.1 General Navier-Stokes equations

The most comprehensive and accurate description of fluid motion is provided by the Navier-Stokes equations, which are a system of nonlinear partial differential equations (PDEs). These equations are derived from the fundamental principles of conservation of mass, momentum, and energy.

Conservation of Mass:

The continuity equation expresses the conservation of mass in differential form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.1)$$

where, ρ is the fluid density, and $\mathbf{u}$ is the fluid velocity vector in 3D.

Conservation of Momentum:

The momentum equation, derived from Newton's second law, along with the assumption that the stress in the fluid is the sum of a diffusing viscous term and a pressure term is:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F} \quad (2.2)$$

where:

- p is the pressure,
- $\mathbf{F}$ represents body forces such as gravity,
- $\boldsymbol{\tau}$ is the deviatoric (viscous) stress tensor, which for Newtonian fluids such as water and steam, as in our case, it's typically taken to be:

$$\boldsymbol{\tau} = \mu [\nabla \mathbf{u} + \nabla \mathbf{u}^T] - \frac{2}{3} \mu \mathbf{I} (\nabla \cdot \mathbf{u})$$

where μ is the dynamic viscosity, and $\mathbf{I}$ is the identity matrix.

Conservation of Energy:

The energy equation, derived from the first law of thermodynamics, is:

$$\frac{\partial}{\partial t} \left[\rho \left(e + \frac{1}{2} u^2 \right) \right] + \nabla \cdot \left[\rho \left(h + \frac{1}{2} u^2 \right) \mathbf{u} \right] = \nabla \cdot (k \nabla T) + \boldsymbol{\tau} \nabla \mathbf{u} + \mathbf{F} \cdot \mathbf{u} + \Phi \quad (2.3)$$

where:

- e is the specific internal energy,
- $h = e + \frac{p}{\rho}$ is the specific enthalpy,

- k is the thermal conductivity,
- T is the temperature,
- $\tau \nabla \mathbf{u}$ represents viscous dissipation,
- Φ is an external heat source.

To close this system of equations (2.1, 2.2, 2.3), we need knowledge of all the unknowns: $\mathbf{u}$, p , ρ , $h \equiv e$, T , μ , and k . The parameters μ and k are typically known from experiments. This leaves us with five basic unknowns ($\mathbf{u}$, p , ρ , h , and T), which can be related by an equation of state (Section 2.1.1). We also need to know the boundary conditions for these unknowns to fully solve the system.

2.1.1 Equation of state: Stiffened Gas

For water under high pressure, the Stiffened Gas model proposed by Harlow and Amsden [7] is often used:

$$\begin{cases} p = (\gamma - 1)\rho e - \gamma p_0, \\ e = \frac{C_p}{\gamma}T - \frac{p_0}{\rho} \end{cases} \quad (2.4)$$

where the parameters (γ , p_0 , C_p) are constants that depend on the material. These parameters can be determined by taking three reference points relevant to our regime.

2.2 Incompressible Navier-Stokes

The system of equations (2.1, 2.2, 2.3) can be further simplified in cases where the variation of the density is very small. When the density is constant and the viscosity μ is also constant, the energy equation (2.3) becomes decoupled from the mass and momentum equations (2.1, 2.2). We can then express the 3D incompressible viscous Navier-Stokes equations, as follows:

$$\begin{cases} \nabla \cdot \mathbf{u} = 0, \\ \rho \frac{\partial \mathbf{u}}{\partial t} + \rho \nabla \cdot [\mathbf{u} \otimes \mathbf{u}] = -\nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{u} + \mathbf{F}. \end{cases} \quad (2.5)$$

2.2.1 Mach number

The Mach number is a dimensionless parameter that measures the ratio of the flow velocity to the speed of the pressure waves in the fluid. It quantifies the compressibility of the flow and is defined as:

$$M = \frac{|\mathbf{u}|}{c_s} \quad (2.6)$$

where c_s is the speed of sound in the medium, and $|\mathbf{u}|$ here denotes the maximum magnitude of the flow velocity.

An important result from Schochet [13] addresses the convergence speed of the compressible regime to an incompressible regime as the Mach number decreases:

Assume that $M \ll 1$, the initial data is close to an incompressible regime $\rho = \rho_*$, $p = p_*$, $\mathbf{u} = \mathbf{u}_*$ with the following asymptotic conditions:

$$\begin{cases} \mathbf{u}(\vec{x}, t = 0) = \mathbf{u}_*(\vec{x}, t = 0) + \mathcal{O}(M)(\vec{x}) \\ p(\vec{x}, t = 0) = p_* + \mathcal{O}(M^2)(\vec{x}) \\ \rho(\vec{x}, t = 0) = \rho_* + \mathcal{O}(M)(\vec{x}) \end{cases}$$

where $\mathbf{u}_*$ is a divergence free vector field ($\nabla \cdot \mathbf{u}_* = 0$), ρ_* is a constant density and p_* a constant pressure.

Then, the solution $(u, p, \rho)(\vec{x}, t)$ to the compressible Navier-Stokes equations is close to an incompressible regime:

$$\begin{cases} \mathbf{u}(\vec{x}, t) = \tilde{\mathbf{u}}(\vec{x}, t) + \mathcal{O}(M)(\vec{x}, t) \\ p(\vec{x}, t) = p_* + \mathcal{O}(M^2)(\vec{x}, t) \\ \rho(\vec{x}, t) = \rho_* + \mathcal{O}(M)(\vec{x}, t) \end{cases} \quad (2.7)$$

where $\tilde{\mathbf{u}}$ is the solution of the incompressible Navier-Stokes equations with initial data $\tilde{\mathbf{u}}(\vec{x}, 0) = \mathbf{u}_*(\vec{x})$.

From (2.7), we observe that the compressible pressure is closer (order M^2) to the incompressible regime value than the other two variables, density and velocity (order M). Thus, for general low Mach number flows ($M \ll 1$), an incompressible solution can be used to approximate a compressible solution. This is what we used as a reference in the test cases for this internship.

2.2.2 Reynolds number

The Reynolds number is a dimensionless parameter that predicts the flow pattern, it's defined as :

$$Re = \frac{\rho |\mathbf{u}| D_h}{\mu} \quad (2.8)$$

where, D_h is the hydraulic diameter (a characteristic length). For a flow in a pipe, D_h is the diameter of the pipe.

The Reynolds number measures the relative importance of viscous effects in an incompressible fluid. A low Reynolds number ($Re \ll 1$) indicates that viscous forces are dominant, leading to laminar flow characterized by smooth, constant fluid motion. In contrast, a high Reynolds number ($Re \gg 1$) suggests turbulent flow, where inertial forces dominate, and the flow becomes chaotic and effectively inviscid. The cases encountered in this internship are of high Reynolds number, as we will see later.

2.3 Flow regime inside the reactor vessel

The geometry of the reactor vessel, as shown in figure 1.3, can be approximated as a vertical cylinder with a height of 4.2 m and a diameter of 1.4 m. For the purpose of this simulation, we assume a 1D flow domain with a length $L = 4.2$ m.

We consider a vertical upward flow within the vessel and assume that the flow is steady and the temperature is constant throughout the domain. As a result, we can neglect the energy equation (2.3) in our analysis.

Our analysis considers two distinct configurations based on our objectives:

- **Single channel:** Initially, we treat the entire interior of the reactor as a single channel with an effective hydraulic diameter of 1.4 m.
- **Two parallel channels:** Subsequently, we refine our model by focusing on the fuel assemblies, simulating two parallel channel flows. Here, the hydraulic diameter corresponds to the spacing between individual fuel rods within the assemblies, approximately 1 cm.

The nominal conditions within a reactor vessel are typically as follows:

- **Velocity:** $u = 1$ m/s
- **Pressure:** $p = 1.55 \times 10^7$ Pa
- **Temperature:** $T = 566$ K

Using data from the International Association for the Properties of Water and Steam (IAPWS) [5, 6], we can determine the following fluid properties at the specified pressure and temperature:

- **Density:** $\rho = 740$ Kg/m³
- **Dynamic viscosity:** $\mu \approx 9 \times 10^{-5}$ Pa · s
- **Speed of sound:** $c_s = 1009$ m/s

2.3.1 Flow regime analysis

The Mach and Reynolds numbers will be used to determine the flow regime, as discussed in Sections 2.2.1 and 2.2.2.

For our nominal conditions, the Mach number, calculated using (2.6), is:

$$M = \frac{u}{c_s} = \frac{1}{1009} \approx 10^{-3} \ll 1,$$

Since $M \ll 1$, we are in a low Mach number regime. This allows us to approximate the flow as an incompressible flow.

The Reynolds number, calculated using (2.8), for the following two scenarios:

- Single channel:

For the single channel model with a characteristic length $D_h = 1.4$ m:

$$Re = \frac{\rho u D_h}{\mu} = \frac{740 \times 1 \times 1.4}{9 \times 10^{-5}} \approx 10^7,$$

- Two Parallel channels:

For the two parallel channels model with a characteristic length $D_h = 0.01$ m:

$$Re = \frac{\rho u D_h}{\mu} = \frac{740 \times 1 \times 0.01}{9 \times 10^{-5}} \approx 8 \times 10^4,$$

In both cases, we observe that $Re \gg 1$, indicating that the flow is highly turbulent, this allows us to treat the flow as inviscid.

Given these analyses, we conclude that the flow within the reactor vessel can be approximated using the incompressible inviscid regime.

2.3.2 1D steady inviscid incompressible flow

Under the assumptions of incompressibility and inviscid flow, the 1D steady Navier-Stokes equations simplifies significantly. The governing system (2.5) reduces to:

$$\begin{cases} \frac{\partial u}{\partial x} = 0, \\ \rho \frac{\partial u^2}{\partial x} + \frac{\partial p}{\partial x} = F. \end{cases} \quad (2.9)$$

The first equation in (2.9) implies a constant velocity along the axial direction, which can be set to $u(x) = u_{in}$ where u_{in} is the velocity at the inlet, serving as a boundary condition for the system. Given the assumptions, the system of equations (2.9) simplifies to:

$$\begin{cases} u(x) = u_{in}, \\ \frac{\partial p(x)}{\partial x} = F. \end{cases} \quad (2.10)$$

The system of equations (2.10) is the one we use to obtain the analytical solution for the verification test cases.

It remains necessary to specify that this approach offers a solution with a margin of error given by the low Mach number assumption (2.7). The margin of error between our approach and the real compressible regime is then:

$$\begin{cases} u_{\text{comp}}(x) = u_{\text{incomp}}(x) + \mathcal{O}(M)(x), \\ p_{\text{comp}}(x) = p_{\text{incomp}}(x) + \mathcal{O}(M^2)(x). \end{cases} \quad (2.11)$$

The left-hand side represents the relative error between the real life compressible regime and our incompressible approximation.

System (2.10) is the system that we solve analytically for the various external forces, and system (2.11) gives us the margin of error.

Chapter 3

Studied cases

In this chapter, we aim to verify that the software can accurately simulate the effects of basic forces applied to fluid flow and determine the precision of these simulations. This verification is crucial because it helps us understand potential reasons for failure in more complex cases. For example, if a particular force is not accurately captured according to the verification test, it might explain why a more complex simulation fails.

Therefore, we test how well the gravity, friction forces (both regular and singular), and porosity are captured. We derive for each test a reference solution, and a methodology (section 3.1) to evaluate the error between the numerical and exact solutions.

3.1 Error evaluation

In order to evaluate the accuracy of the numerical simulation, we assess the relative error between the numerical solution f_{code} and the theoretical solution $f_{\text{analytical}}$. More precisely, for a physical quantity f (e.g. pressure or velocity), we define the pointwise relative error $\delta f(x)$ at each point x in the domain as follows:

$$\delta f(x) = \frac{f_{\text{code}}(x) - f_{\text{analytical}}(x)}{f_{\text{analytical}}(x)}. \quad (3.1)$$

In order to measure the discrepancy between $f_{\text{code}}(x)$ the values of the variable computed by the code and $f_{\text{analytical}}(x)$ the proposed analytical solution, we use the **L[∞]-norm** which is defined as:

$$\Delta f = \|\delta f(x)\|_{\infty} = \max \{|\delta f(x)|\} \quad \text{for all } x \text{ in the domain.} \quad (3.2)$$

However in practice, we do not know the compressible analytical solution $f_{\text{analytical}}$. We use instead the incompressible analytical solution $f_{\text{inc, analytical}}$, which is easier to find and is close to $f_{\text{analytical}}$. Indeed, the discrepancy between $f_{\text{inc, analytical}}$ and $f_{\text{analytical}}$ is given in equation (2.11) and we have

$$\delta f(x) = \frac{f_{\text{code}}(x) - f_{\text{inc, analytical}}(x)}{f_{\text{inc, analytical}}(x)} + \frac{f_{\text{inc, analytical}}(x) - f_{\text{analytical}}(x)}{f_{\text{analytical}}(x)}. \quad (3.3)$$

For low Mach flows, the first term is usually main the contribution in the error (3.3). It can be evaluated using the code output f_{code} and the incompressible solution $f_{\text{inc, analytical}}$.

Once the first term of the relative error in (3.3) has been estimated, we should complement it with an estimate of the second term $\frac{f_{\text{inc, analytical}}(x) - f_{\text{analytical}}(x)}{f_{\text{analytical}}(x)}$. From (2.11) we can deduce that this second term is of order M when f is the velocity and M^2 if f is the pressure.

We should then define a threshold for the acceptability of the numerical solution f_{code} . The Mach number M being small and dimensionless, it seems natural to define a threshold based on M . The second contribution to the relative error (3.3) is of order M in the worst case. Therefore a sufficient condition for $\Delta f \approx \mathcal{O}(M)$ is simply $\frac{f_{\text{code}}(x) - f_{\text{inc, analytical}}(x)}{f_{\text{inc, analytical}}(x)} \approx \mathcal{O}(M)$.

We therefore propose to accept the numerical result if the following condition is satisfied:

$$\Delta f = \frac{f_{\text{code}}(x) - f_{\text{inc,analytical}}(x)}{f_{\text{inc,analytical}}(x)} < M \quad (3.4)$$

for f being the velocity unknown and then f being pressure unknown.

In practice, after running the simulation, we compute $\frac{f_{\text{code}}(x) - f_{\text{inc,analytical}}(x)}{f_{\text{inc,analytical}}(x)}$ at each point in the domain, and then take the maximum absolute value across all points. We note here, that to avoid having infinity when $f_{\text{inc,analytical}}(x) = 0$, we artificially add a very small number $\epsilon = 10^{-10}$ to the denominator.

This framework provides a basis for evaluating whether the simulation accurately captures the physical phenomena.

3.2 Single channel

3.2.1 Case 1.1: 1D single phase channel under gravity

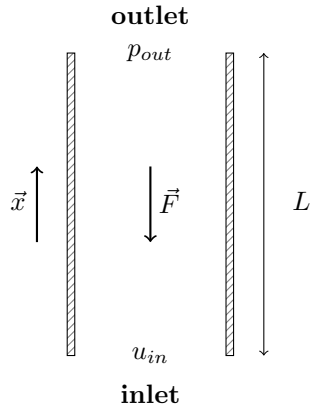


Figure 3.1: Illustration of case 1.1.

In this case, we aim to verify that the software can accurately simulate the effects of gravity on fluid flow and determine the precision of this simulation using our error evaluation scheme proposed in Section 3.1. To achieve this, we propose an analytical solution for both velocity and pressure, which will be incorporated into the verification test cases of the software. The test has been implemented using a Jupyter Notebook and then converted to a PDF, and the details can be found in Appendix A.1.

Test description: To isolate the impact of gravity, we simplify the physical model by averaging or ignoring other physical aspects. The reactor vessel is approximated as a single channel with vertical upward flow. We assume no horizontal variations, indicating that the flow is purely one-dimensional and vertical. Viscosity is neglected, the temperature is constant, and the only acting force is gravity, represented by:

$$\vec{F} = \rho \vec{g}, \quad (3.5)$$

where $\|\vec{g}\| = 9.81 \text{ m/s}^2$ is the gravitational acceleration of the Earth, and ρ is the density, which is free to vary in the software.

This setup is illustrated in figure 3.1. We consider a domain $x \in [0, L]$ with the following boundary conditions:

- **Inlet:** $u(0) = u_{in}$
- **Outlet:** $p(L) = p_{out}$
- **Walls:** Adiabatic (i.e., no heat exchange) and no-slip ($(\vec{u} \cdot \vec{n} = 0)$, i.e., the velocity at the wall is zero)

Fixing the initial flow speed u_{in} is analogous to a pump inside the reactor, and fixing the outlet pressure p_{out} , is analogous to a pressurizer.

Simulation: The software simulates this case using the above description. While we do not present the full initialization details, we briefly outline the setup:

- A 1x1x50 mesh grid is employed, consisting of 50 vertical cells,
- The initial velocity u_{in} is set only in the vertical direction to (0,0,1) m/s,
- The outlet pressure is set to $p_{out} = 1.55 \times 10^7$ Pa,
- The gravitational force is set to $(0, 0, -\rho g)$, where ρ is initially set to 740 kg/m^3 , and is then free to vary,
- The simulation operates in the steady (or stationary) regime, implying that velocity and pressure will not vary with time.

The software simulates the real compressible regime using its own internal model. At the end of the simulation, we retrieve three quantities: the velocity in the vertical direction $u(x)$, the pressure $p(x)$, and the Mach number $M(x)$. These are used to evaluate the error using the evaluation scheme presented in Section 3.1, and for this, we need the analytical solution.

Analytical solution: As demonstrated in Section 3.1, we are dealing with a low Mach, highly turbulent flow, which allows us to approximate our flow regime to an incompressible inviscid regime. For this case, we can approximate it in 1D, in the steady regime. We then use the 1D steady incompressible system given by (3.8), which reduces to solving the following set of equations:

$$\begin{cases} u(x) = u_{in}, \\ \frac{\partial p(x)}{\partial x} = -\rho g. \end{cases}$$

Note that the density ρ is constant here. Integrating for $x \in [0, L]$, and with $p(L) = p_{out}$, gives us:

$$\begin{cases} u(x) = u_{in}, \\ p(x) = p_{out} - \rho g(x - L). \end{cases} \quad (3.6)$$

where:

$$u_{in} = 1 \text{ m/s}, \quad p_{out} = 1.55 \times 10^7 \text{ Pa}, \quad \rho = 740 \text{ kg/m}^3, \quad g = 9.81 \text{ m/s}^2, \quad L = 4.2 \text{ m}.$$

Comparison of numerical and incompressible solutions: Comparison: After running the simulation with the code, we retrieve the velocity and pressure profiles and plot them against the above analytical solution in figure 3.2.

We also retrieve the Mach number $M(x)$ from the simulation, which, for this case, is practically constant, $M(x) = 0.001 \equiv M$.

We then compute the maximum absolute relative error across all points of the domain, using (3.4) for velocity and pressure; we get:

$$\begin{aligned} \Delta u &= 0.00004 < 0.001 \\ \Delta p &= 0.0000015 < 0.001 \end{aligned}$$

Since $M = 0.001$, both Δu and Δp satisfy the condition of being less than M .

In figure 3.4, we observe a very satisfactory agreement between the analytical and simulated pressure profiles. The velocity profile shows some discrepancies at first glance; however, this is primarily due to the assumption of constant density in the analytical solution, whereas the software allows for density variations. Despite this, the relative error in the velocity remains well below the threshold we proposed. Thus, we conclude that the simulation of single-phase upward flow under the force of gravity is validated.

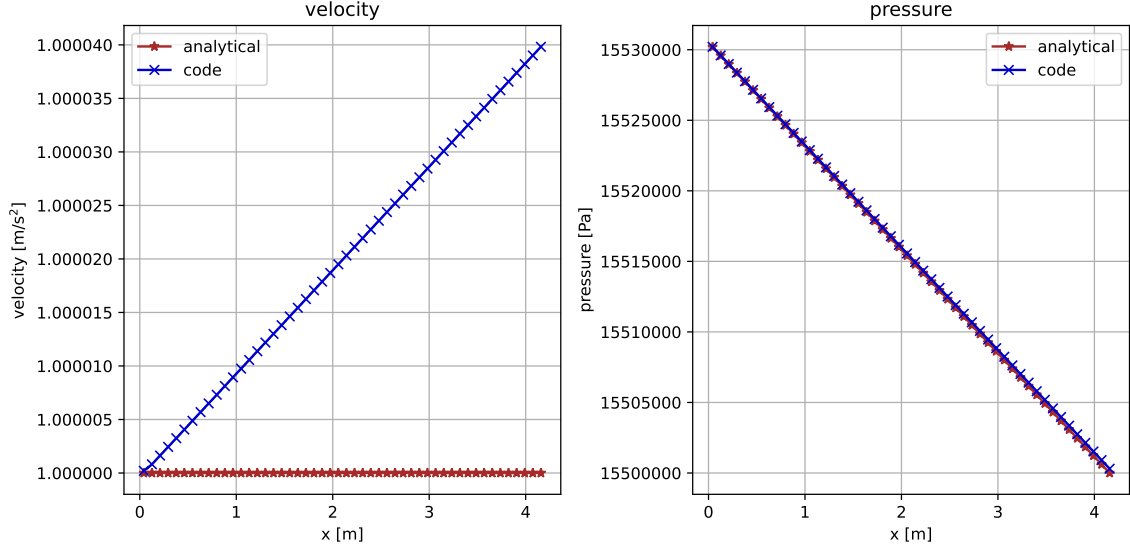


Figure 3.2: case 1.1 velocity and pressure profiles.

3.2.2 Case 1.2: 1D single phase channel with regular friction (pressure loss)

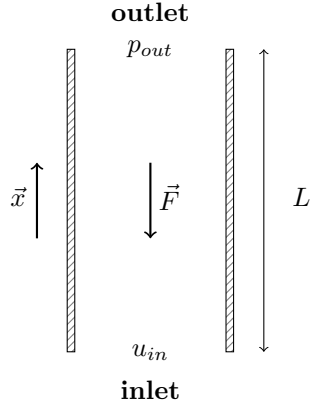


Figure 3.3: Illustration of case 1.2.

In this case, we verify that the software can accurately simulate the effects of friction on fluid flow and determine the precision of this simulation using our error evaluation scheme proposed in Section 3.1.

Test description: This test case examines fluid flow through a vertical channel where friction is the dominant force, rather than gravity as in Case 1.1. The reactor vessel is again approximated as a single channel with upward flow, and we assume no horizontal variations, making the flow purely one-dimensional and vertical. Here, we neglect viscosity and temperature changes, focusing solely on the frictional force, which we will model using the pressure loss equation:

$$\vec{F} = -\frac{C_f}{2D_h} \rho(p) \|\vec{u}\| \vec{u} \quad (3.7)$$

where C_f is the Darcy friction factor, which depends on the flow regime and geometry of the pipe. It is usually between 0.01 and 10; however, for verification purposes, we want to highlight the pressure loss effect, so we set $C_f = 100$. D_h , as before, is the hydraulic diameter and is set to 1.4 m. The density initially is $\rho = 740 \text{ kg/m}^3$ and is free to vary in the software.

This setup is illustrated in figure 3.3. We consider a domain $x \in [0, L]$ with the same boundary conditions as the Case 1.1:

- **Inlet:** $u(0) = u_{in}$
- **Outlet:** $p(L) = p_{out}$
- **Walls:** Adiabatic and no-slip ($\vec{u} \cdot \vec{n} = 0$)

Simulation: The software simulates the case similarly to Case 1.1, using the same setup for the mesh and initial conditions. We only change the force acting: we set gravity to zero and replace it with the pressure loss equation defined above with a constant friction factor $C_f = 100$.

Analytical solution: As in Case 1.1, we use the 1D steady incompressible system given by Equation (3.8), which reduces to solving the following set of equations:

$$\begin{cases} u(x) = u_{in}, \\ \frac{\partial p(x)}{\partial x} = -\frac{C_f}{2D_h} \rho |u(x)| u(x). \end{cases}$$

Note that the density ρ is constant here. Integrating for $x \in [0, L]$, and with $p(L) = p_{out}$, gives us:

$$\begin{cases} u(x) = u_{in}, \\ p(x) = p_{out} - \frac{C_f}{2D_h} \rho u_{in}^2 (x - L). \end{cases} \quad (3.8)$$

where:

$$u_{in} = 1 \text{ m/s}, \quad p_{out} = 1.55 \times 10^7 \text{ Pa}, \quad \rho = 740 \text{ kg/m}^3, \quad C_f = 100, \quad L = 4.2 \text{ m}, \quad D_h = 1.4 \text{ m}.$$

Comparison of numerical and incompressible solutions: After running the simulation with the code, we retrieve the velocity and pressure profiles and plot them against the above analytical solution in figure 3.4.

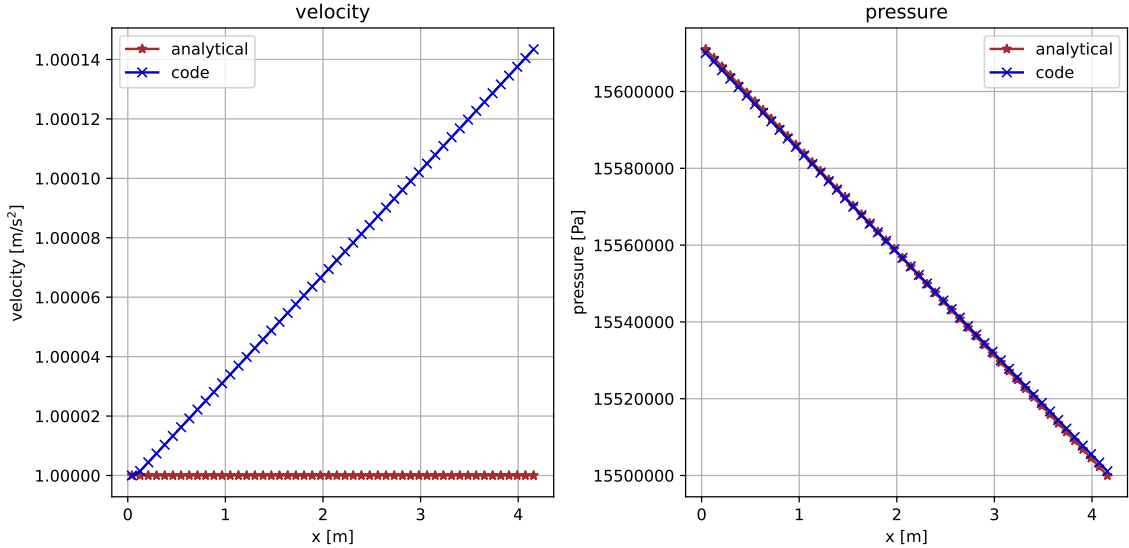


Figure 3.4: case 1.2 velocity and pressure profiles.

We also retrieve the Mach number $M(x)$ from the simulation, which, for this case, is also practically constant, $M(x) = 0.001 \equiv M$.

We then compute the maximum absolute relative error across all points of the domain, using (3.4) for velocity and pressure; we get:

$$\begin{aligned} \Delta u &= 0.00014 < 0.001 \\ \Delta p &= 0.0000072 < 0.001 \end{aligned}$$

Since $M = 0.001$, both Δu and Δp satisfy the condition of being less than M .

In figure 3.4, we observe behavior similar to Case 1.1: a very good agreement in the pressure profiles and some discrepancies in the velocity profile due to the assumption of constant density in the analytical solution. Despite this assumption, the relative error in the velocity remains well below the threshold we proposed.

Initially, we ignored viscosity, but it is effectively reintroduced through the external force of friction, which we modeled as a pressure loss.

Thus, we conclude that the simulation of single-phase upward flow subject to regular friction is validated.

3.2.3 Case 1.3: 1D single phase channel with singular friction

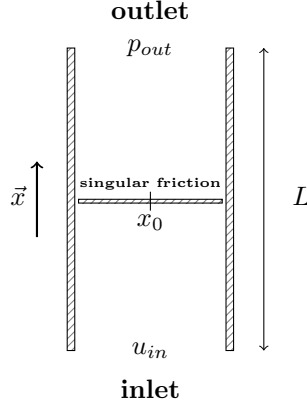


Figure 3.5: Illustration of case 1.3.

In this case, we verify that the software can accurately simulate the effects of singular friction on fluid flow and determine the precision of this simulation using our error evaluation scheme proposed in Section 3.1.

Test description: This test case examines fluid flow through a vertical channel where singular friction is introduced at a specific point within the domain. The reactor vessel is again approximated as a single channel with upward flow, and we assume no horizontal variations, making the flow purely one-dimensional and vertical. Here, we neglect viscosity and temperature changes. Singular friction is introduced at $x = x_0 \in (0, L)$, modeled using the pressure loss equation:

$$\vec{F} = -\frac{C_f}{2D_h}\delta(x - x_0)\rho(p)||\vec{u}||\vec{u} \quad (3.9)$$

where $\delta(x - x_0)$ is the Dirac delta function at $x = x_0$. The density is free to vary in the software.

This setup is illustrated in figure 3.5. We consider a domain $x \in [0, L]$ with the same boundary conditions as the Case 1.1:

- **Inlet:** $u(0) = u_{in}$
- **Outlet:** $p(L) = p_{out}$
- **Walls:** Adiabatic and no-slip ($\vec{u} \cdot \vec{n} = 0$)

Simulation: The software simulates the case similarly to Case 1.1, using the same setup for the mesh and initial conditions. Gravity is set to zero, and the only acting force is the singular friction equation (3.9) with a constant friction factor $C_f = 100$, $D_h = 1.4$ m, $x_0 = 2.1$ m, and density initially set to $\rho = 740$ kg/m³.

Analytical solution: As in Case 1.1, we use the 1D steady incompressible system given by (3.8), which reduces to solving the following set of equations:

$$\begin{cases} u(x) = u_{in}, \\ \frac{\partial p(x)}{\partial x} = -\frac{C_f}{2D_h}\delta(x - x_0)\rho||u(x)||u(x). \end{cases}$$

Note that the density ρ is constant here. The pressure equation becomes:

$$\frac{d}{dx}p(x) = -\frac{\rho C_f u_{in}^2}{2D_h}\delta(x - x_0)$$

Integrating the Dirac delta function gives us the Heaviside step function $H(x - x_0) + \text{constant}$, where:

$$H(x - x_0) = \begin{cases} 0, & \text{if } x < x_0 \\ 1, & \text{if } x \geq x_0 \end{cases}$$

Thus, the pressure becomes,

$$p(x) = -\frac{\rho C_f u_{in}^2}{2D_h}H(x - x_0) + \text{constant} \quad (3.10)$$

To determine the constant, we evaluate the pressure at boundary $x = L > x_0$, where we set $p(L) = p_{out}$

$$\begin{aligned} p(L) \equiv p_{out} &= -\frac{\rho C_f u_{in}^2}{2D_h} \times 1 + \text{constant}, \\ \Rightarrow \text{constant} &= p_{out} + \frac{\rho C_f u_{in}^2}{2D_h} \end{aligned}$$

Plugging this back in (3.10), gives us:

$$p(x) = p_{out} - \frac{\rho C_f u_{in}^2}{2D_h}(H(x - x_0) - 1) \quad (3.11)$$

Finally the velocity and pressure for this case can be expressed as:

$$\begin{aligned} u(x) &= u_{in}, \\ p(x) &= \begin{cases} p_{out} + \frac{\rho C_f u_{in}^2}{2D_h} & \text{if } 0 \leq x < x_0, \\ p_{out} & \text{if } x_0 \leq x \leq L. \end{cases} \end{aligned} \quad (3.12)$$

where:

$$\begin{aligned} u_{in} &= 1 \text{ m/s}, \quad p_{out} = 1.55 \times 10^7 \text{ Pa}, \quad \rho = 740 \text{ kg/m}^3, \quad C_f = 100, \\ x_0 &= 2.1 \text{ m} \quad L = 4.2 \text{ m}, \quad D_h = 1.4 \text{ m}. \end{aligned}$$

Comparison of numerical and incompressible solutions: After running the simulation with the code, we retrieve the velocity and pressure profiles and plot them against the above analytical solution in figure 3.6.

We also retrieve the Mach number $M(x)$ from the simulation, which, for this case, is also practically constant, $M(x) = 0.001 \equiv M$.

We then compute the maximum absolute relative error across all points of the domain, using (3.4) for velocity and pressure; we get:

$$\begin{aligned} \Delta u &= 0.00005 < 0.001 \\ \Delta p &= 0.000002 < 0.001 \end{aligned}$$

Since $M = 0.001$, both Δu and Δp satisfy the condition of being less than M .

In figure 3.6, we can clearly see the effect of singular friction on our results. At $x_0 = 2.1 \text{ m}$, there is a significant jump in pressure as computed by the code and expected analytically. Physically, this pressure loss is analogous to a sudden change in the geometry of the pipe, such as a narrow pipe connected to a reservoir. The results demonstrate that our approach successfully captures the correct behavior with an acceptable error.

Thus, we conclude that the simulation of single-phase upward flow subject to singular friction is validated.

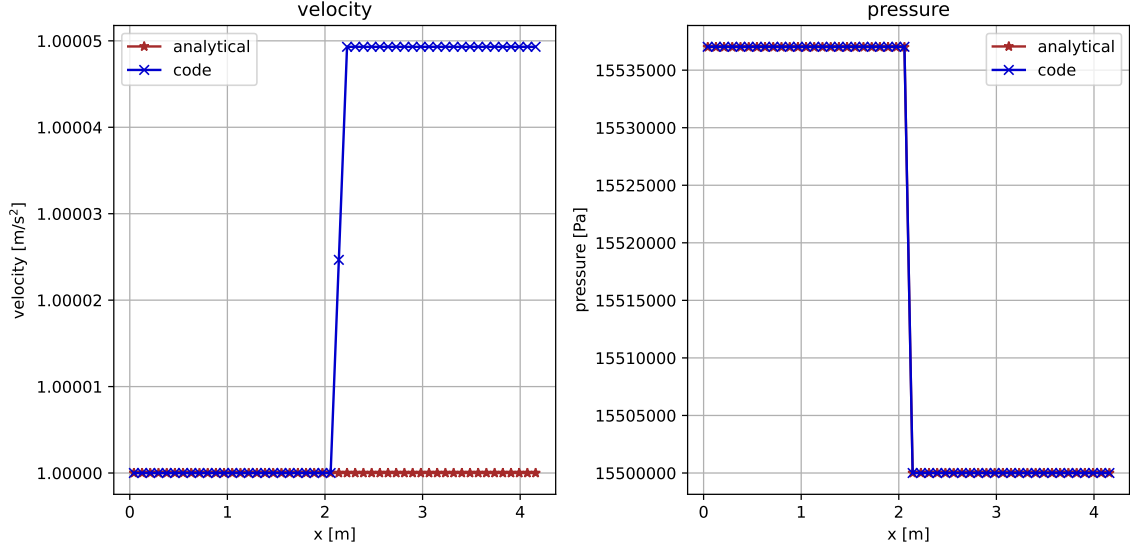


Figure 3.6: case 1.3 velocity and pressure profiles.

3.3 Two channels

3.3.1 Case 2: Two parallel channels subject to regular friction

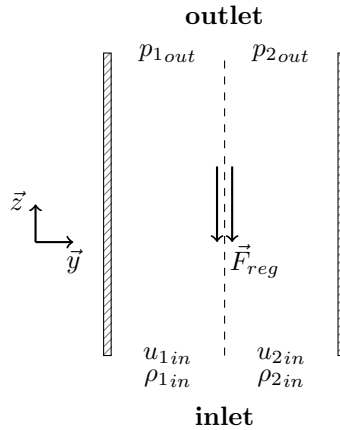


Figure 3.7: Illustration of case 2.

Test description: This test case involves simulating fluid flow through two adjacent parallel channels with no solid separation between them, allowing for potential mixing. The dynamics of such systems have not been thoroughly explored in previous studies. This new case will be added to the verification database.

In the context of a nuclear reactor vessel, the vessel typically contains about 150 to 250 fuel assemblies, each consisting of approximately 200 to 300 fuel rods. Figure 1.4 shows a typical fuel assembly and rods, We approximate each rod to be 1 x 1 cm in cross-section and 4.2 m in length. The water flows through the gaps between the rods, evacuating heat generated by the nuclear fuel (e.g., uranium) inside the rods. The gap between the rods is typically about 1 cm.

This case focuses on simulating two such parallel flows, verifying that the software can capture the effects of mixing between the two channels.

Simulation: The setup for this case is illustrated in figure 3.7. We consider a domain split into two channels:

- **Channel 1:** $y \in [0, \frac{L_y}{2}]$, $z \in [0, L_z]$
- **Channel 2:** $y \in [\frac{L_y}{2}, 0]$, $z \in [0, L_z]$

with $L_y = 0.02$ m and $L_z = 4.2$ m. Viscosity and gravity are neglected, the temperature is constant (no heating). The external force applied is regular friction, represented by:

$$\vec{F}_{reg} = -\frac{C_f}{2D_h}\rho(p)||\vec{u}||\vec{u}$$

where C_f is the Darcy friction factor, which we take to be 0.02. The hydraulic diameter $D_h = 0.01$ m is the gap between the fuel rod.

The following boundary conditions are:

- **Inlet 1:** at $z = 0$, $0 < y < \frac{L_y}{2}$; $\vec{u}_1 = (0, u_{1in} = 2.7 \text{ m/s})$ and $\rho_1 = \rho_{1in} = 740 \text{ kg/m}^3$
- **Inlet 2:** at $z = 0$, $\frac{L_y}{2} < y < L_y$; $\vec{u}_2 = (0, u_{2in} = 1.5 \text{ m/s})$ and $\rho_2 = \rho_{2in} = 740 \text{ kg/m}^3$
- **Outlet 1:** at $z = L_z$, $0 < y < \frac{L_y}{2}$; $p_1 = p_{1out} = 1.55 \times 10^7 \text{ Pa}$
- **Outlet 2:** at $z = L_z$, $\frac{L_y}{2} < y < L_y$; $p_2 = p_{2out} = 1.55 \times 10^7 \text{ Pa}$
- **Outer walls:** at $y = 0$, $y = L_y$; Adiabatic and no-slip ($\vec{u} \cdot \vec{n} = 0$)
- **Inner boundary:** at $y = \frac{L_y}{2}$; no physical wall

The software simulates the real compressible regime for the case with the above description. At the end of the simulation, we retrieve the velocity and density profiles for channel 1, $u_1(z)$, and for channel 2, $u_2(z)$, ρ_2 . These results are then compared to the analytical solution, which we will derive below.

Analytical solution: We want to derive an analytical solution for the two parallel channels. For this case, we take the following approximations:

- Almost incompressible single-phase fluid

$$M = \frac{||\vec{u}||}{c_s} \ll 1. \quad (3.13)$$

- Upward flow along the z axis:

$$u_z \gg u_y \text{ and } L_z \gg L_y. \quad (3.14)$$

In particular $u_z > 0$.

- No heat exchange, i.e the temperature is constant.
- variations according to y are small compared to variations according to z :

$$|u_y \partial_y u_z| \ll |u_z \partial_z u_z|. \quad (3.15)$$

- Isobaric (consequence of the two previous ones, to be quantified mathematically) :

$$\partial_y P = 0. \quad (3.16)$$

This is in fact a consequence of (3.15) and of the assumption

$$|\partial_z u_y| \ll 1,$$

(see Appendix A.2).

- Regular friction (pressure drop) :

$$\vec{F}_{reg} = -K\rho||\vec{u}||\vec{u}. \quad (3.17)$$

with $K = \frac{C_f}{2D_h}$, due to the presence of solid obstacles within the flow.

- No gravity or viscosity

$$\vec{g} = 0, \quad \mu = 0. \quad (3.18)$$

- Identical outlet pressure for both channels

$$p_{1out} = p_{2out}. \quad (3.19)$$

- Identical inlet density for both channels

$$\rho_{1in} = \rho_{2in}. \quad (3.20)$$

We seek particular approximate solutions to the 2D compressible Navier-Stokes equations for a fluid subject to a regular friction force $\vec{F}_{reg} = -K\rho\|\vec{u}\|\vec{u}$.

The general compressible Navier-Stokes equations for the mass, momentum, and energy conservation are given by (2.1, 2.2, and 2.3) respectively. In this case, viscosity is neglected and temperature is constant, thus the energy equation is decoupled from the first two. The 2D steady inviscid compressible system that we will solve is:

$$\nabla \cdot (\rho \vec{u}) = 0 \quad (3.21)$$

$$\nabla \cdot [\rho \vec{u} \otimes \vec{u}] + \nabla p = -K\rho\|\vec{u}\|\vec{u}, \quad (3.22)$$

The system in Cartesian coordinates in the (y, z) plane:

$$\partial_y(\rho u_y) + \partial_z(\rho u_z) = 0 \quad (3.23)$$

$$\partial_y(\rho u_y^2) + \partial_z(\rho u_y u_z) + \partial_y p = -K\rho\|\vec{u}\|u_y \quad (3.24)$$

$$\partial_y(\rho u_y u_z) + \partial_z(\rho u_z^2) + \partial_z p = -K\rho\|\vec{u}\|u_z. \quad (3.25)$$

In each channel we have a set of 2D stationary Navier-Stokes equations :

- Channel 1

$$\begin{cases} \partial_y(\rho_1 u_{1y}) + \partial_z(\rho_1 u_{1z}) = 0 \\ \partial_y(\rho_1 u_{1y}^2) + \partial_z(\rho_1 u_{1y} u_{1z}) + \partial_y p_1 = -K\rho_1\|\vec{u}_1\|u_{1y} \\ \partial_y(\rho_1 u_{1y} u_{1z}) + \partial_z(\rho_1 u_{1z}^2) + \partial_z p_1 = -K\rho_1\|\vec{u}_1\|u_{1z}. \end{cases} \quad (3.26)$$

- Channel 2

$$\begin{cases} \partial_y(\rho_2 u_{2y}) + \partial_z(\rho_2 u_{2z}) = 0 \\ \partial_y(\rho_2 u_{2y}^2) + \partial_z(\rho_2 u_{2y} u_{2z}) + \partial_y p_2 = -K\rho_2\|\vec{u}_2\|u_{2y} \\ \partial_y(\rho_2 u_{2y} u_{2z}) + \partial_z(\rho_2 u_{2z}^2) + \partial_z p_2 = -K\rho_2\|\vec{u}_2\|u_{2z}. \end{cases} \quad (3.27)$$

We take the difference between the two z -momentum equations of systems (3.26) and (3.27):

$$\begin{aligned} & \partial_y(\rho_1 u_{1y} u_{1z}) + \partial_z(\rho_1 u_{1z}^2) + \partial_z p_1 = -K\rho_1\|\vec{u}_1\|u_{1z} \\ - \\ & \partial_y(\rho_2 u_{2y} u_{2z}) + \partial_z(\rho_2 u_{2z}^2) + \partial_z p_2 = -K\rho_2\|\vec{u}_2\|u_{2z} \end{aligned}$$

and get:

$$\partial_y(\rho_1 u_{1y} u_{1z} - \rho_2 u_{2y} u_{2z}) + \partial_z(\rho_1 u_{1z}^2 - \rho_2 u_{2z}^2) + \partial_z(p_1 - p_2) = -K(\rho_1\|\vec{u}_1\|u_{1z} - \rho_2\|\vec{u}_2\|u_{2z}). \quad (3.28)$$

Now from:

- the isobaric assumption (3.16) we deduce $\partial_z(p_1 - p_2) \approx 0$.
- the mass balance (3.23)

$$\partial_y(\rho_1 u_{1y} u_{1z}) = \rho_1 u_{1y} \partial_y u_{1z} + u_{1z} \partial_y(\rho_1 u_{1y}) \quad (3.29)$$

$$= \rho_1 u_{1y} \partial_y u_{1z} - u_{1z} \partial_z(\rho_1 u_{1z}). \quad (3.30)$$

From the small y derivatives assumption (3.15), (3.30) yields

$$\partial_y(\rho_1 u_{1y} u_{1z}) \approx -u_{1z} \partial_z(\rho_1 u_{1z}),$$

and the almost incompressible assumption (3.13) yields

$$\partial_y(\rho_1 u_{1y} u_{1z}) = -\frac{1}{2} \partial_z(\rho_1 u_{1z}^2). \quad (3.31)$$

- the vertical flow assumption (3.14) we get $\|\vec{u}_1\| u_{1z} \approx u_{1z}^2$ and $\|\vec{u}_2\| u_{2z} \approx u_{2z}^2$.

Hence (3.28) is roughly equivalent to

$$\frac{1}{2} \partial_z (\rho_1 u_{1z}^2 - \rho_2 u_{2z}^2) = -K (\rho_1 u_{1z}^2 - \rho_2 u_{2z}^2). \quad (3.32)$$

Taking $g(z) = \rho_1 u_{1z}^2 - \rho_2 u_{2z}^2$, the differential equation (3.32) can then be expressed as:

$$\int_0^z \frac{dg(z)}{g(z)} = \int_0^z -2K dz \quad (3.33)$$

We can solve it as follows:

$$\begin{aligned} \ln(|g(z)|) - \ln(|g(z=0)|) &= -2Kz \\ \ln(|g(z)|) &= \ln(|g(z=0)|) - 2Kz \\ |g(z)| &= e^{-2Kz} |g(z=0)| \end{aligned} \quad (3.34)$$

plugging back the value of $g(z)$:

$$|\rho_1 u_{1z}^2 - \rho_2 u_{2z}^2| = e^{-2Kz} |\rho_1 u_{1z}^2 - \rho_2 u_{2z}^2|_{z=0}. \quad (3.35)$$

$$(3.36)$$

We multiply by $\frac{1}{2}$ to make appear a kinetic energy term; and using $K = \frac{C_f}{2D_h}$. Finally, we get:

$$\left| \frac{1}{2} \rho_1 u_{1z}^2 - \frac{1}{2} \rho_2 u_{2z}^2 \right| = e^{-\frac{C_f}{D_h} z} \left| \frac{1}{2} \rho_{1in} u_{1in}^2 - \frac{1}{2} \rho_{2in} u_{2in}^2 \right| \quad (3.37)$$

where:

$$\begin{aligned} u_{1in} &= 2.7 \text{ m/s}, \quad u_{2in} = 1.5 \text{ m/s}, \quad p_{out} = 1.55 \times 10^7 \text{ Pa}, \quad \rho_{1in} = \rho_{2in} = 740 \text{ kg/m}^3, \\ C_f &= 0.02, \quad D_h = 0.01 \text{ m}. \end{aligned}$$

We can see here that we weren't able to derive explicit equations for the velocities, density, or pressure, as doing so would require further assumptions. Instead, we derived Equation (3.37), which predicts that the difference in kinetic energy density between the two channels will decrease exponentially with the term $e^{-\frac{C_f}{D_h} z}$.

This is an important result that we will use to determine if the software agrees with our approach. We also note that the evaluation of the error between the analytical and numerical results will differ in this case since we don't have explicit formulas for the velocity or pressure. Instead, we propose an alternative evaluation method below.

Comparison of numerical and analytical solutions: After running the simulation, we retrieve the velocity and density profiles for the two channels and compute the quantity:

$$\Delta KE = \left| \frac{1}{2} \rho_1 u_{1z}^2 - \frac{1}{2} \rho_2 u_{2z}^2 \right| \quad (3.38)$$

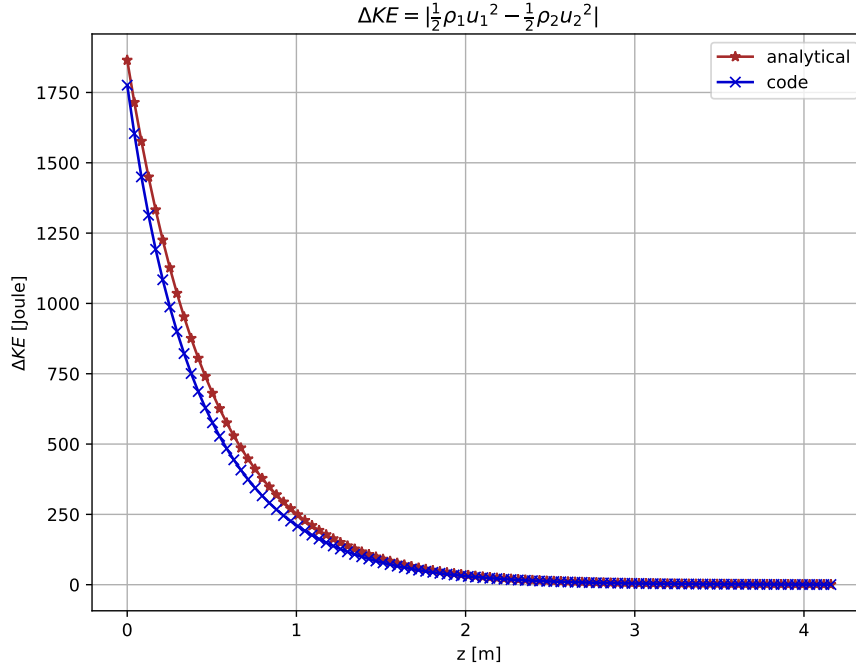


Figure 3.8: Difference in kinetic energy density the two parallel channels (case 2).

We then compare this to the analytical solution (3.37).

Figure 3.8 shows the comparison between the numerical and analytical results. We have a very promising result; the two curves align nicely. We can see how the difference in kinetic energy decreases asymptotically to finally reach zero, indicating that there is no longer any difference in kinetic energy between the two channels, i.e., the two channels become homogeneous and indistinguishable. This is an encouraging result as it aligns with the expected physical behavior of this case, as we can clearly see the "mixing length" phenomena.

To quantify the results, we evaluate the error and decide if it is acceptable.

Here we compute the maximum and average relative errors between the numerical and analytical results and obtain:

- maximum relative error = 0.1642
- average relative error = 0.0927

At first, this may seem high, especially compared to previous cases. However, it makes sense upon closer inspection, since we don't have an explicit function for a single variable as before. Instead, we have a function combining four variables: ρ_1 , u_1^2 , ρ_2 , and u_2^2 . Thus, the error will naturally be higher. We propose using the error propagation method to determine the acceptable threshold for this case.

Error propagation

Given the function:

$$f = AB^2 - CD^2$$

we want to calculate the propagated uncertainty in f using the method of partial derivatives. The propagated uncertainty formula is:

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial A}\sigma_A\right)^2 + \left(\frac{\partial f}{\partial B}\sigma_B\right)^2 + \left(\frac{\partial f}{\partial C}\sigma_C\right)^2 + \left(\frac{\partial f}{\partial D}\sigma_D\right)^2}$$

where:

$$\begin{aligned}\frac{\partial f}{\partial A} &= B^2, & \frac{\partial f}{\partial B} &= 2AB, \\ \frac{\partial f}{\partial C} &= -D^2, & \frac{\partial f}{\partial D} &= -2CD\end{aligned}$$

The propagated uncertainty becomes:

$$\sigma_f = \sqrt{(B^2\sigma_A)^2 + (2AB\sigma_B)^2 + (D^2\sigma_C)^2 + (2CD\sigma_D)^2}$$

In our case, for (3.38), we get

$$\sigma_{\Delta KE} = \sqrt{(u_1^4\sigma_{\rho_1})^2 + (2\rho_1 u_1^2\sigma_{u_1})^2 + (u_2^4\sigma_{\rho_2})^2 + (2\rho_2 u_2^2\sigma_{u_2})^2} \quad (3.39)$$

We can estimate the uncertainty σ for the velocities and densities based on Section 2.2.1. Since the relative error on u and ρ is of the order of $M = 0.001$, we propose to take the uncertainty of the variable to be $10^{-1}M$ multiplied by the maximum value of the variable, such that:

$$\begin{aligned}\sigma_{\rho_1} &= 10^{-4} \times \max(\rho_1), \\ \sigma_{u_1} &= 10^{-4} \times \max(u_1), \\ \sigma_{\rho_2} &= 10^{-4} \times \max(\rho_2), \\ \sigma_{u_2} &= 10^{-4} \times \max(u_2)\end{aligned} \quad (3.40)$$

By doing this, we can compute the uncertainty of ΔKE at each point and from it, compute the theoretical relative error at each point:

$$\text{theoretical relative error} = \frac{\sigma_{\Delta KE}}{\Delta KE} \quad (3.41)$$

Finally, we take the average of (3.41) and obtain 0.156, a theoretical proposed threshold for this case.

Threshold	0.156
Maximum actual relative error	0.1642
Average actual relative error	0.0927

Table 3.1: Error distribution on ΔKE

Table 3.1 summarizes the error distribution for the two parallel channels test. The actual errors are within the range of the proposed threshold, which means that the software does indeed capture the behavior of mixing of two channels.

Thus, we conclude that the simulation of two channels with single-phase upward flow under regular friction is validated.

Chapter 4

Conclusion and perspectives

This internship project aimed to enhance the verification of numerical simulations of fluid dynamics in nuclear reactor systems. The primary objectives were to develop an error evaluation scheme for single-channel cases, introduce a new test case for two parallel channels, and improve the documentation of the verification database using Jupyter Notebook.

1. Error Evaluation Scheme:

- We successfully introduced a scientifically backed error evaluation scheme for 1D single-channel, single-phase flows.
- The software results were validated against this scheme, confirming its effectiveness and reliability.

2. Improved Documentation:

- We improved the documentation of the verification database by rewriting test cases in Jupyter Notebook. This approach allows for easier modification and automatic generation of PDF reports.
- Beyond the cases detailed in this report, we worked on integrating all tests present in the database into a more user-friendly format. An example of a test case documented in Jupyter Notebook and converted into a PDF can be found in Appendix A.1.

3. Two-Channel Flow Case:

- We successfully addressed the complex scenario of two parallel channels with different initial velocities by deriving an analytical solution, which had not been previously accomplished.
- An evaluation scheme specific to this test was proposed, ensuring the accurate representation of the mixing phenomena and determination of the mixing length.
- This new test case will be added to the verification database, expanding its coverage and applicability.

The outcomes of this internship significantly contribute to the robustness of the verification process for thermohydraulic simulation codes at CEA.

Perspectives This internship laid the ground for further works on the verification database of thermohydraulics simulations.

1. Single Phase Single Channel: Further single phase tests can be upgraded with the new methodology for the verifying the numerical results (porosity jumps, 2D tests ...).
2. Two Phase Single Channel: Extend the new verification methodology to two phase flows. This involves a clear definition of the Mach(s) number(s) involved in a two flows, as well as an convergence results towards the incompressible regime.
3. Single Phase Two Channels: Add more two channels cases based on singular friction, viscosity and heat conduction.

4. Two Phase Two Channels: investigate whether it is possible to extend the current methodologies to tackle two phase two channels flow.
5. Define a framework for the sharing of the base Jupyter Notebooks between several software. The base Jupyter Notebook should be located in the common open source kernel used by all the software. The technical difficulty is that every software should be able to complement the base Jupyter Notebook with the output of its simulations to form the final verification Notebook.

Bibliography

- [1] Histoire de la création du cea. <https://www cea.fr/Pages/le-cea/histoire-creation-CEA.aspx>. Accessed: 2024-05-17.
- [2] The discovery of nuclear fission. <https://www.mpic.de/4469988/die-entdeckung-der-kernspaltung#:~:text=Nuclear%20fission%20was%20discovered%20at,also%20created%20in%20the%20process>. Accessed: 2024-05-23.
- [3] Active and Passive Nuclear Safety. URL <https://www.nuclear-power.com/nuclear-power/reactor-physics/nuclear-safety/safety-systems/active-and-passive-nuclear-safety/>. Accessed: 2024-05-17.
- [4] Country nuclear profile summary - france. <https://cnpp.iaea.org/public/countries/FR/profile/highlights>. Accessed: 2024-05-23.
- [5] International Association for the Properties of Water and Steam. IAPWS R12-08. Release on the iapws formulation 2008 for the viscosity of ordinary water substance, 2008. URL <http://www.iapws.org/relguide/viscosity.html>.
- [6] International Association for the Properties of Water and Steam. IAPWS R7-97(2012). Revised release on the iapws industrial formulation 1997 for the thermodynamic properties of water and steam, 2012. URL <http://www.iapws.org/relguide/IF97-Rev.html>.
- [7] F. Harlow and A. Amsden. Fluid dynamics. Technical Report LA-4700, Los Alamos National Laboratory, 1971.
- [8] Institut de Radioprotection et de Sûreté Nucléaire (IRSN). Composants d'un réacteur. URL <https://www.irsn.fr/savoir-comprendre/surete/composants-dun-reacteur-detail>. Accessed: 2024-05-27.
- [9] IRSN. Fonctionnement d'un réacteur nucléaire. URL <https://www.irsn.fr/savoir-comprendre/surete/fonctionnement-dun-reacteur-nucleaire>. Accessed: 2024-05-27.
- [10] MikeRun. Own work, 2017. URL <https://commons.wikimedia.org/w/index.php?curid=60375907>. CC BY-SA 4.0.
- [11] U.S. nrc. The pressurized water reactor, 2017. URL <https://www.nrc.gov/reading-rm/basic-ref/students/animated-pwr.html>.
- [12] Oriol Planas. Pressurized water reactor (PWR): working and pros and cons, 2017. URL <https://nuclear-energy.net/nuclear-power-plants/nuclear-reactor/types/pressurized-water-reactor>.
- [13] S. Schochet. Fast singular limits of hyperbolic pdes. *Journal of Differential Equations*, 114(2): 476–512, 1994. ISSN 0022-0396. doi:<https://doi.org/10.1006/jdeq.1994.1157>. URL <https://www.sciencedirect.com/science/article/pii/S0022039684711570>.
- [14] TEPCO. URL <https://www.tepco.co.jp/en/hd/about/facilities/nuclear-e.html>. Accessed: 2024-08-04.
- [15] Union of Concerned Scientists. A Brief History of Nuclear Accidents Worldwide, 2013. URL <https://www.ucsusa.org/resources/brief-history-nuclear-accidents-worldwide>.

Appendix A

Appendix

A.1 Generated report using Jupyter notebook for case 1.1

We present the generated PDF for the single channel single phase flow test case discussed in Section 3.2.1. It's important to note that general details on how to run the numerical simulations, such as discretizations, solvers, problem initialization, etc. that you can see in this PDF were not presented in this report. However, we still worked on them during the internship.

1 Treset 1.1.1: Écoulement 1D monophasique avec gravité

1.1 Introduction

- Validation made by: Ali Fadlallah
- Report generated: 05/08/2024

Ce document s'inscrit dans le cadre d'une campagne de vérification des discrétisations/ solveurs disponibles dans le code dans le code de thermohydraulique. Cette vérification est basée sur des comparaisons à des solutions analytiques. Il s'agit ici du test 1.1.1 de cette campagne, test pour lequel nous disposons d'une solution analytique.

Le test est décrit au travers des chapitres suivants :

- Introduction
- Description du cas
- Solutions numériques
- Solutions analytiques
- Bilan des résultats
- Évaluation de l'erreur
- Conclusion

1.2 Description du cas

Ce cas simule un écoulement monophasique dans un canal avec une force de gravité. Les parois sont adiabatiques et le fluide utilisé est de l'eau, considéré parfait (pas de pertes de charge).

Les conditions aux limites à l'entrée sont de type vitesse & enthalpie, et à la sortie du type pression.

Les dimensions du canal sont indiquées ci-dessous.

1.2.1 Données géométriques

Géométrie	Valeur
Longueur X [m]	1.4
Largeur Y [m]	1.4
Hauteur Z [m]	4.2
Diamètre hydraulique Dh [m]	1.4

1.2.2 Maillage

Mailles	Valeur
nb mailles Z	50
dz [m]	0.084
nb mailles X	1
dx [m]	1.4
nb mailles Y	1
dy [m]	1.4

1.2.3 Données physiques

Fluide	Eau
Gravité [m.s ⁻²]	(0, 0, -9.81)
Densité [kg.m ⁻³]	740

1.2.4 Conditions limites

Entrée	Valeur
Enthalpie liquide [J/kg]	1.3e+06
Enthalpie vapeur [J/kg]	2.6e+06
Vitesse liquide [m.s ⁻¹]	(0, 0, 1)
Vitesse vapeur [m.s ⁻¹]	(0, 0, 0)

Sortie	Valeur
Pression [Pa]	155e+05

1.2.5 Schémas numériques testés

On teste 4 schémas en alternant entre deux types de discrétisation (VDF/PolyMAC) et deux types de solveurs (jacobienne_complete/solveur_pression):

Cases	Discrétisation	Solveur
PolyMAC_jc	PolyMAC	jacobienne_complete
PolyMAC_sp	PolyMAC	solveur_pression
VDF_jc	VDF	jacobienne_complete
VDF_sp	VDF	solveur_pression

1.3 Solutions numériques

Running ****with Sserver****...
=> A total of 4 cases were (successfully) run in 5.1s.

1.4 Solutions analytiques

La solution analytique suivante concerne un écoulement incompressible soumis à une force de gravité $F = -\rho g$.

Les grandeurs d'intérêt de ce test sont la vitesse massique et la pression, dont les évolutions le long du canal sont tracées ci-dessous.

$$u(z) = u_{liq}^{in},$$

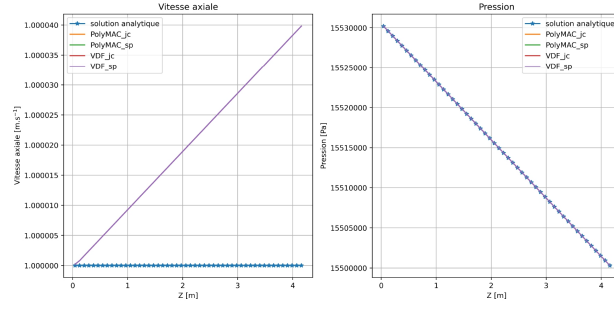
$$P(z) = P_{out} - g\rho_{liq}^{in}(z - L),$$

avec :

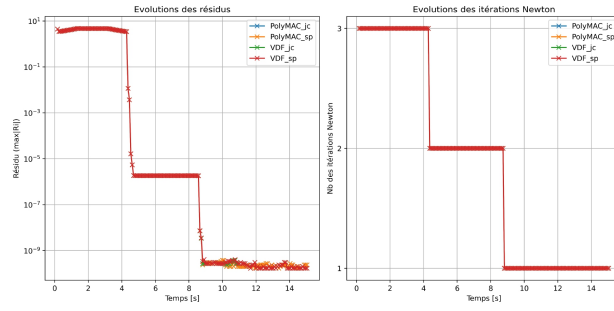
$$u_{liq}^{in} = 1 \text{ m.s}^{-1}, \quad P_{out} = 155 \times 10^5 \text{ Pa}, \quad \rho_{liq}^{in} = 740 \text{ kg.m}^{-3},$$

$$g = 9.81 \text{ m.s}^{-2}, \quad L = 4.2 \text{ m}$$

1.5 Résultats



1.5.1 Convergences



1.6 Évaluation des Erreurs

Les résultats obtenus avec le code sont en principe pour un écoulement compressible. Nous proposons de les comparer à la solution analytique incompressible monophasique en introduisant le nombre de Mach M , qui est le rapport entre la vitesse de l'écoulement et la vitesse du son dans le liquide, soit $\frac{u}{c}$. La solution compressible $(u, p)(\vec{x}, t)$ est proche de la solution incompressible $(u_*, p_*)(\vec{x}, t)$ comme suit :

$$\begin{cases} u(\vec{x}, t) = u_*(\vec{x}, t) + \mathcal{O}(M)(\vec{x}, t) \\ p(\vec{x}, t) = p_*(\vec{x}, t) + \mathcal{O}(M^2)(\vec{x}, t) \end{cases}$$

Cela signifie que l'erreur relative entre la vitesse u compressible (code) et incompressible (solution analytique) sera proportionnelle à M , et celle de la pression p à M^2 .

On trouve une valeur constante de $M \approx 10^{-3}$ pour tous les cas testés (voir tableau ci-dessous)

	PolyMAC_jc	PolyMAC_sp	VDF_jc	VDF_sp
Mach number	0.001	0.001	0.001	0.001

Deux critères sont considérés pour que le test soit réussi.

- **Erreur relative:** L'erreur relative entre la solution numérique calculée par le code thermohydraulique et la solution analytique correspondante est calculée comme suit :

$$\text{Erreur \%} = \max \left| \frac{V_{\text{code}} - V_{\text{analytique}}}{V_{\text{analytique}}} \right| \times 100$$

où V_{code} est la valeur calculée par code thermohydraulique, $V_{\text{analytique}}$ est la valeur analytique.

Les critères de succès sont définis en fonction du nombre de Mach :

- **Pression :** Le critère de succès est fixé à $100 \times M^2 = 10^{-4}$, soit 0.01%.
- **Vitesse massique :** Le critère de succès est fixé à $M = 10^{-3}$, soit 0.1%.
- **Ecart de variation P_{in} - P_{out} :** L'écart entre la variation $P_{\text{in}} - P_{\text{out}}$ calculée par le code et la solution analytique est calculé comme suit :

$$\Delta P_{\text{in-out}} = \left| \left[\frac{P_{\text{in}} - P_{\text{out}}}{P_{\text{in}}} \right]_{\text{code}} - \left[\frac{P_{\text{in}} - P_{\text{out}}}{P_{\text{in}}} \right]_{\text{analytique}} \right| \times 100$$

Le critère de succès est fixé à 1%.

Le tableau suivant présente les différentes erreurs pour chaque cas test :

- ✓ est affiché si l'erreur est acceptable
- ! est affiché si l'erreur est dans les 5 % du critère.
- ✗ est affiché si l'erreur dépasse le critère.

Une fiche est vérifiée, c'est à dire sans anomalie, si aucune ligne ne contient ✗ comme résultat.

cases	ΔP	ΔP_{in-out}	Δu
PolyMAC_jc	0.002%✓	0.002%✓	0.004%✓
PolyMAC_sp	0.002%✓	0.002%✓	0.004%✓
VDF_jc	0.002%✓	0.002%✓	0.004%✓
VDF_sp	0.002%✓	0.002%✓	0.004%✓

Les résultats sont satisfaisants.

1.7 Conclusion

Cette fiche est vérifiée avec succès.

A.2 Isobaric assumption

In order to quantify the estimate $\partial_y p \approx 0$, we take the difference between the two y -momentum equations of systems (3.26) and (3.27) that we recall hereafter:

$$\partial_y(\rho_1 u_{1y}^2) + \partial_z(\rho_1 u_{1y} u_{1z}) + \partial_y p_1 = -K \rho_1 \|\vec{u}_1\| u_{1y} \quad (\text{A.1})$$

$$\partial_y(\rho_2 u_{2y}^2) + \partial_z(\rho_2 u_{2y} u_{2z}) + \partial_y p_2 = -K \rho_2 \|\vec{u}_2\| u_{2y} \quad (\text{A.2})$$

we find that $\partial_y(p_1 - p_2)$ is related to u_{1y} and u_{2y} and their derivatives:

$$\partial_y(p_1 - p_2) = \partial_y(\rho_2 u_{2y}^2 - \rho_1 u_{1y}^2) + \partial_z(\rho_2 u_{2y} u_{2z} - \rho_1 u_{1y} u_{1z}) - K(\rho_1 \|\vec{u}_1\| u_{1y} - \rho_2 \|\vec{u}_2\| u_{2y}) \quad (\text{A.3})$$

From the mass balance (3.23)

$$\partial_z(\rho_1 u_{1y} u_{1z}) = \rho_1 u_{1z} \partial_z u_{1y} + u_{1y} \partial_z(\rho_1 u_{1z}) \quad (\text{A.4})$$

$$= \rho_1 u_{1z} \partial_z u_{1y} - u_{1y} \partial_y(\rho_1 u_{1y}). \quad (\text{A.5})$$

We find that $\partial_y p \approx 0$ under the assumptions

$$\begin{aligned} |u_{1y}| &\ll 1, & |u_{2y}| &\ll 1 \\ |\partial_z u_{1y}| &\ll 1, & |\partial_z u_{2y}| &\ll 1. \end{aligned} \quad (\text{A.6})$$